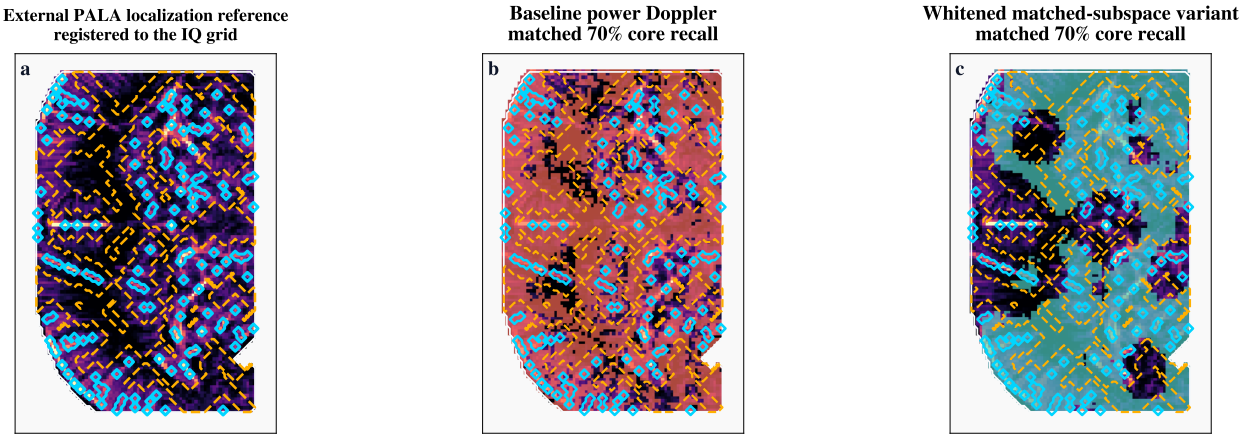
Localized Matched-Subspace Detection for Functional Ultrasound and Ultrafast Doppler Imaging

Arthur Shune*
SkyMesa Systems Inc.

Abstract

Functional ultrasound is now used for intraoperative, neonatal, and mobile brain imaging, but even after standard clutter suppression, tissue motion, phase instability, and aliased flow can produce false detections that distort vascular maps and contaminate functional readouts. We study a family of localized matched-subspace detection statistics applied after conventional clutter suppression of beamformed fUS slow-time data, ranging from a fixed band-limited statistic to locally whitened variants. On the held-out SIMUS structural benchmark, the fixed variant cuts nuisance false-positive rate from 0.998 to 0.004 at matched recall 0.5, eliminating structured artifact leakage on the same clutter-filtered residual under the benchmark operating point. On one open real-IQ rat-brain dataset, the fully whitened variant shows a consistent structural improvement across all 10 audited blocks (exact paired sign-flip test on block-mean AUC, $p = 0.002$), improving AUC from 0.504 for power Doppler to 0.530 on a conservative same-acquisition vessel-core versus shell endpoint anchored by a published localization-derived vascular reference. These results demonstrate that the downstream score head is a separable, consequential design choice in fUS pipelines, and that band-limited matched-subspace scoring reduces nuisance leakage on the same clutter-filtered residual; broader real-data functional validation remains future work.



*Correspondence: arthur@skymesasystems.com. ORCID: 0009-0004-6433-3856.

Figure 1: Localization-derived structural-label result on ULM Zenodo 7883227 using the published PALA rat-brain localization rendering as an external vascular reference. Left: the PALA-derived vessel-core and perivascular-shell masks after registration back to the IQ grid. Middle and right: the same audited 64-frame MC–SVD residual window (block 001, frames 128:192), thresholded at matched 70% vessel-core recall for baseline power Doppler and the prespecified no-registration whitened matched-subspace variant. Table 1 reports the formal same clutter-filtered-residual comparison of scoring functions.

Score	AUC (H1 vs H0)	TPR@10^{-2}	TPR@10^{-3}	shell FPR @70% core recall
Baseline (power Doppler)	0.504 [0.498,0.510]	0.010 [0.009,0.011]	7.9e-5 [0,2.4e-4]	0.698 [0.689,0.706]
Baseline (Kasai lag-1 magnitude)	0.503 [0.496,0.509]	0.009 [0.008,0.010]	5.9e-5 [0,1.8e-4]	0.701 [0.693,0.709]
Fixed matched-subspace head	0.514 [0.508,0.521]	0.014 [0.013,0.016]	2.6e-4 [1.2e-4,4.3e-4]	0.703 [0.694,0.711]
Adaptive head	0.514 [0.507,0.520]	0.014 [0.013,0.016]	2.6e-4 [1.2e-4,4.3e-4]	0.702 [0.693,0.711]
Fully whitened variant	0.530 [0.524,0.537]	0.024 [0.021,0.027]	0.004 [0.003,0.006]	0.676 [0.665,0.687]

Table 1: Same-residual ULM structural audit on real in vivo rat-brain IQ. All rows use the same frozen 64-frame MC–SVD residual cube, the same localization-derived vessel-core versus shell masks, and the same SCM whitening recipe where applicable; only the downstream score head changes. Entries report window-level means with 95% bootstrap intervals over the 70 audited windows. The fully whitened variant exceeds baseline power Doppler on all 10 audited blocks (exact paired sign-flip test on block-mean AUC, $p = 0.002$).

1 Introduction

Functional ultrasound (fUS) is now used for vascular and functional mapping in settings ranging from preclinical neuroscience to awake surgery and cranial-window human imaging. In these settings, the practical problem is often not average sensitivity alone, but that even a few artifactual false detections can distort maps and undermine interpretation. Tissue motion, phase instability, and aliased high-velocity flow therefore matter most when they survive ordinary clutter suppression and populate the extreme right tail of the detector output.

In intra-operative settings, a false-positive activation region can mislead surgical planning. In resting-state connectivity studies, a single artifactual hot spot that survives across blocks can create a spurious seed correlation. In neonatal bedside monitoring, where acquisition windows are short and motion is unavoidable, artifact leakage in the Doppler map can be hard to distinguish from genuine hemodynamic variation without independent confirmation that is rarely available. These are not average-sensitivity problems; they are rare-event problems driven by the right tail of the score distribution.

On the held-out SIMUS-Struct benchmark, changing only the final scoring function eliminates all nuisance detections at matched 50% target-flow recall on the same clutter-filtered residual: on the intra-operative setting, nuisance FPR falls from 0.998 for Kasai to 0.004 for the fixed matched-subspace detector, and the corresponding fixed chain eliminates all nuisance detections on all four held-out seed-setting pairs. That is the headline result of the paper.

The method is intentionally narrow. After ordinary clutter suppression, it asks a local question: within a small neighborhood, is the remaining slow-time signal concentrated in a prespecified flow band, or is it more consistent with structured clutter residual? We study three downstream scoring functions on that fixed residual stream: a fixed matched-subspace detection statistic, an adaptive variant that selectively invokes local slow-time whitening, and a fully whitened variant that applies

whitening throughout. Among these, the fixed statistic is the default on the held-out structural benchmark, while stronger whitening helps in narrower clutter-heavy or structure-concentration settings. Figure 10 later distills that practical choice into a label-free deployment rule.

The scope is equally narrow. This paper is about downstream detection-statistic design after a fixed clutter filter, not about proposing another clutter filter. The same clutter-filtered residual comparisons are the point: PD, Kasai, whitened power, and matched-subspace variants are evaluated on identical residual cubes so scoring-function effects can be separated from clutter-filter effects.

The evidence proceeds from held-out structural benchmarking to stricter mechanism checks and bounded real-IQ corroboration. We use the short labels kWave-Stress, Gammex-Struct, SIMUS-Struct-Mobile, SIMUS-Struct-Intraop, ULM-CoreShell, ULM-Motion, and Shin-ProxyMotion throughout; SIMUS-Struct denotes the paired mobile and intra-operative structural benchmark.

This problem matters now because fUS is no longer only a preclinical proof-of-concept modality. It is already used in intra-operative, neonatal, cranial-window, and emerging mobile settings [1, 8, 9, 20–22], where rare nuisance-driven false positives can matter more than average image quality.

2 Background and Problem Formulation

Problem setup. For each spatial location (or tile-aggregated snapshot), let $x \in \mathbb{C}^{L_t}$ denote a slow-time vector. We consider

$$H_0 : \; x = c + n, \qquad H_1 : \; x = s + c + n,$$

where s is flow-consistent signal, c is structured clutter/motion, and n is residual noise. In practice, H_0 is spatially nonstationary and heavy-tailed, motivating local covariance adaptation and tail-focused evaluation.

Core notation and bands. We use band operators (P_o, P_f, P_g, P_a) for clutter/DC, flow, guard, and alias regions, respectively. Physically, P_f is the Doppler band where the targeted microvascular flow is expected, P_g is a transition guard band, and P_a captures aliased high-velocity flow or vibration. For short ensembles, these are exact orthogonal DFT-bin projectors; for longer apertures they are leakage-controlled analysis operators and are treated as band-energy statistics with explicit sanity sentinels (Supplementary Material).

2.1 Related context

Spatiotemporal SVD, RPCA, HOSVD, and local adaptive clutter filtering are established in ultrafast Doppler and fUS, and learned clutter filters are relevant prior work on the residual-formation side of the pipeline [11, 12, 30, 31, 44]. These methods improve the residual itself; none explicitly evaluates the downstream detection statistic as a separable design choice on the same residual. The question here is narrower: once a clutter-filtered slow-time cube has been fixed, can the downstream detection statistic reduce artifact leakage more effectively than PD- or Kasai-style readouts? That separates this paper from Yu-, Baranger-, and Song-style local clutter filtering, which adapts the clutter filter itself before a conventional Doppler score is applied [13, 14, 16]. ULM motion correction, twinkling-phantom structure audits, and hemodynamic fUS readouts therefore enter here as evaluation context and downstream audits rather than as the primary algorithmic contribution; the fuller literature context is kept in the methods companion.

2.2 Baseline scores and detector definitions

Beamformed slow-time data. Let $X[t, y, x] \in \mathbb{C}$ denote a beamformed slow-time IQ movie over $t = 0, \ldots, T-1$ and spatial indices (y, x). We first apply a baseline clutter-suppression operator (e.g. motion-compensated SVD), producing a residual cube $X_{\text{base}}[t, y, x]$ used for both baseline scoring and as the input to the detector family.

Power Doppler proxy and baseline scores. From the baseline-filtered cube we define the baseline ROC score map

$$S_{\text{base}}(y, x) := \frac{1}{T} \sum_{t=0}^{T-1} |X_{\text{base}}(t, y, x)|^2 . \tag{1}$$

This is ordinary power Doppler on X_{base}. For short-ensemble settings we additionally report auxiliary conventional comparators such as log-power Doppler and Kasai lag-1 magnitude [15] so that baseline-score idiosyncrasies do not dominate the comparison.
When the same power statistic is used only as a display or proxy map for deterministic preprocessing, we write

$$\text{PD}^{\uparrow}(y, x) \equiv S_{\text{base}}(y, x),$$

to make it explicit that the quantity is being used for visualization or mask construction rather than as the ROC score itself. S_{base} denotes the baseline ROC score map on X_{base}, S_{det} the matched-subspace detector score map, and $\text{PD}^{\uparrow}$ a display or proxy map used only for visualization or deterministic preprocessing rather than for ROC scoring. Unless explicitly stated, the detector family consists of three detector variants built from the same band-limited score family: a fixed matched-subspace head without whitening, a prespecified adaptive variant that promotes selected pixels onto a whitened branch when guard-band evidence indicates clutter, and a fully whitened covariance-adaptive variant.

3 Method

What changes relative to SVD + PD. Relative to a conventional ultrafast Doppler pipeline, the proposed detector family changes the final scoring stage rather than replacing upstream clutter suppression. After forming a residual slow-time signal, we score its energy inside a prespecified flow subspace rather than using only aggregate Doppler power. The simplest fixed head uses that score without local whitening, a prespecified adaptive variant promotes clutter-heavy pixels onto a whitened branch only when guard-band evidence warrants it, and a fully whitened variant is retained for settings where stronger local adaptation is beneficial. The goal is not simply to increase vessel conspicuity, but to reduce nuisance-driven high scores in operating regions where false positives matter.

Plain-language interpretation. The fixed head is a local band-limited flow detector applied after ordinary clutter filtering. The adaptive branch adds tile-local whitening only where guard-band energy indicates localized clutter.

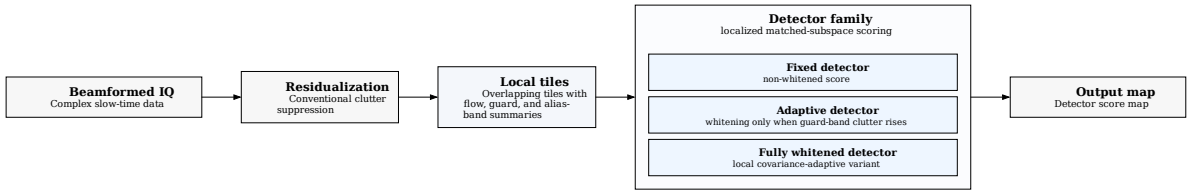


Figure 2: Method overview. Beamformed IQ first undergoes ordinary clutter filtering, after which the detector operates tilewise in slow time. The fixed detector uses the band-limited matched-subspace score everywhere. The adaptive detector uses that same score by default and switches to a whitened branch only in tiles with strong guard-band evidence of localized clutter. The fully whitened detector applies whitening everywhere.

Pipeline overview.

1. Baseline clutter suppression (e.g. MC–SVD) produces a residual cube X_{base} and baseline score $S_{\text{base}}(y, x)$.
2. For each tile Ω_i, form the local slow-time support used by the chosen detector head; when whitening is active, apply the fixed setting-specific local covariance transform $y_{i,k} = R_i^{-1/2} x_{i,k}$.
3. Compute the self-normalized matched-subspace score on the chosen tile representation and aggregate over overlapping tiles to obtain the detector score map. The fixed and fully whitened forms are given explicitly in Eqs. (2) and (3); setting $R_i = I$ recovers the simpler band-limited fixed head.
4. If the adaptive variant is used, retain the fixed head by default and promote pixels onto the whitened branch only where a prespecified guard-band clutter feature exceeds a fixed threshold.

In practical terms, the method asks a local question after ordinary clutter suppression: within a small spatial neighborhood, does the remaining slow-time activity look more like microvascular flow or more like locally structured nuisance? Detector-core ablations on labeled brain simulations and the Gammex phantom are reported in the main Results.

The derivations below formalize the three detector modes introduced above.

Three detector variants. On tile i, let $x_{i,k}$ denote the unwhitened slow-time snapshots and $y_{i,k} = R_i^{-1/2} x_{i,k}$ the whitened snapshots when whitening is active. The fixed detector is

$$S_0(i) := \frac{1}{K} \sum_{k=1}^{K} \frac{x_{i,k}^{\mathrm{H}} P_f \, x_{i,k}}{x_{i,k}^{\mathrm{H}} (I - P_f) \, x_{i,k} + \rho \, x_{i,k}^{\mathrm{H}} x_{i,k} + \epsilon}, \tag{2}$$

which is the same band-limited matched-subspace ratio with $R_i = I$. The fully whitened detector is

$$S_w(i) := \frac{1}{K} \sum_{k=1}^{K} \frac{y_{i,k}^{\mathrm{H}} P_f \, y_{i,k}}{y_{i,k}^{\mathrm{H}} (I - P_f) \, y_{i,k} + \rho \, y_{i,k}^{\mathrm{H}} y_{i,k} + \epsilon}. \tag{3}$$

In plain language, the numerator measures energy in the expected flow band, while the denominator measures energy everywhere else plus a small stabilizing ridge; the ratio is high only when flow-band energy dominates the rest of the slow-time spectrum. The detector family has three concrete

variants. The fixed detector uses $S_0(i)$ everywhere. The prespecified adaptive detector uses

$$S(i) = \begin{cases} S_w(i), & r_g^{\text{base}}(i) \geq \tau_g, \\ S_0(i), & r_g^{\text{base}}(i) < \tau_g, \end{cases} \tag{4}$$

where $r_g^{\text{base}}(i)$ is a baseline guard-band energy fraction and τ_g is fixed once and then held fixed across all reported experiments. The fully whitened detector uses $S_w(i)$ everywhere. In words, the adaptive detector keeps the simpler score in clean tiles and only turns on full whitening where the residual itself shows guard-band evidence of localized clutter.

Tile support and switching rule. All reported detectors operate on overlapping local tiles. In the main quantitative settings, the spatial support is fixed at 8×8 pixels, with slow-time aperture $L_t = 8$ for kWave-Stress, $L_t = 16$ for Gammex-Struct, and $L_t = 64$ for SIMUS-Struct and ULM-CoreShell. The adaptive rule evaluates the baseline guard-band fraction $r_g^{\text{base}}(i)$ on each tile and switches from $S_0(i)$ to $S_w(i)$ only when $r_g^{\text{base}}(i) \geq \tau_g$; otherwise the fixed score is retained. Full details on sliding-window temporal embedding, covariance estimation, and conditioning are kept in the methods companion.
Deterministic label-free integrity checks for proxy construction, band discretization, and conditional execution are documented in the methods companion. Invalid configurations are excluded from reporting, and decision-rule variants such as conditional execution are reported separately from the default detector pipeline.

4 Experimental Design

4.1 Prespecified operating protocol and operating-point reporting

All processing settings (tile geometry, aperture, band definitions, and adaptive threshold) are fixed for each setting before held-out results are reported. Detection is evaluated by thresholding the high-score tail (right tail) at fixed FPR targets:

$$\mathbb{P}\{S \geq \tau_\alpha \mid H_0\} = \alpha, \qquad \text{TPR}(\alpha) = \mathbb{P}\{S \geq \tau_\alpha \mid H_1\}.$$

For each experiment, we report the supported floor $fpr_{\min} = 1/n_{\text{neg}}$ and bootstrap uncertainty at the natural unit of variation (windows or frames, depending on setting). These choices enforce a strict low-FPR operating discipline and avoid over-interpreting unsupported tail points. We emphasize $\alpha = 10^{-3}$–10^{-4} because those operating points expose nuisance-tail failure modes cleanly, not because routine fUS use must literally operate at those exact values. The same map-trustworthiness problem is already visible at a conventional matched background FPR of 10^{-2} in the worked Gammex example of Figure 7, where the baseline thresholded map is dominated by off-lumen detections while the matched-subspace score recovers lumen-consistent structure.

Threshold calibration and deployment. Unless otherwise stated, τ_α is calibrated per window (or per cine sequence) on that unit's negative set to report $\text{TPR}(\alpha)$ at the intended operating point; in labeled settings, the negative set is available for *exact* calibration, while in deployment it would be estimated via a negative-only training procedure or from a fixed calibration set. The Results section therefore includes two non-retrospective transfer checks: a fixed-calibration audit on SIMUS-Struct and a complementary block-split fixed-calibration audit on the ULM structural audit.

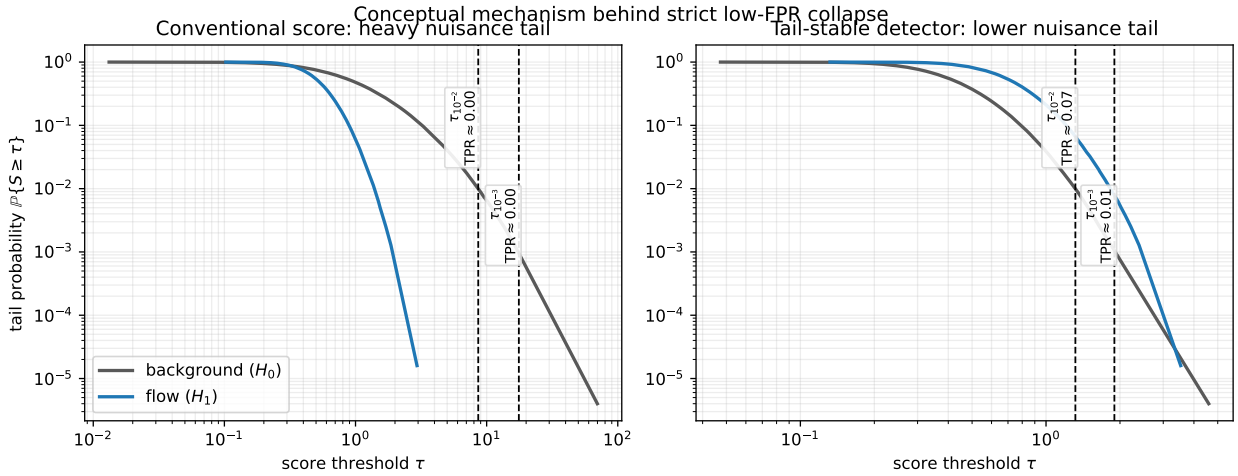


Figure 3: Conceptual strict-tail mechanism. Left: a conventional score can still show moderate-FPR separation while failing at a stricter operating point because the background right tail pushes the background-calibrated threshold $\tau_{10^{-3}}$ above essentially all flow pixels, yielding TPR ≈ 0 by construction. Right: when the nuisance tail is reduced, the same strict background allowance leaves some flow pixels above threshold. The later labeled-brain and Gammex tail figures are dataset-specific examples of this same mechanism.

How the prespecified defaults were chosen. Band edges are tied to acquisition physics, while tile geometry and L_t are chosen once per setting to preserve spatial adaptation with stable detector estimates and then held fixed. The adaptive threshold τ_g is a single frozen guard-energy cutoff chosen to keep the fixed score in cleaner settings and enable whitening only when guard-band clutter evidence is clearly non-negligible; full whitening-estimator and conditioning details are kept in the methods companion.
Additional deterministic integrity checks for acquisition metadata and compute gating, together with their logged reason codes, are listed in the methods companion.

4.2 Regimes and scope

The evidence is stratified by setting so that quantitative detection results remain clearly separated from robustness or transfer audits. Downstream detector-head design matters after fixed residualization; the fixed band-limited head is the default structural choice on the held-out SIMUS benchmark; adaptive and fully whitened variants help in narrower clutter-heavy or structure-concentration settings; and fixed-threshold transfer is encouraging in selected settings but not solved.

- **kWave-Stress.** These are controlled stress tests for strict-tail behavior. Across the open-skull, alias-dominant, and structured-clutter scenarios, the matched-subspace detector family retains measurable sensitivity at ultra-low FPR, with the strongest gains in the alias-dominant and structured-clutter settings.
- **Gammex-Struct.** These provide independent structure-based labels on real scanner data. We derive B-mode-only tube masks, compare conventional short-ensemble Doppler scores with matched-subspace detector-family scores, and report structural low-FPR operating points and

within-ensemble motion ladders.

- **SIMUS-Struct.** This is the main held-out synthetic benchmark in the paper. It uses the SIMUS/PyMUST forward model and is motivated by reported fUS operating conditions rather than by generic simulator coverage alone [8,9,34,45–47]. We report held-out structural and functional performance for the prespecified SIMUS-Struct-Mobile and SIMUS-Struct-Intraop settings, together with a held-out clutter-heavy stress test on SIMUS-Struct-Mobile under cardiac-like pulsation and short ensembles.
- **ULM-CoreShell.** This dataset is used for motion-robustness analyses on real IQ, with controlled rigid and nonrigid motion injection and explicit fairness checks. We also report a structural audit using the published PALA super-resolution vascular rendering registered back onto the IQ grid, with a vessel-core versus perivascular-shell endpoint.
- **Shin-ProxyMotion.** This open real-IQ dataset provides a second label-free motion-stress evaluation under a prespecified low-frequency operating setting.
- **Supplementary analyses.** Additional label-free audits (e.g. PD-only whole-brain fUS) and diagnostic panels are provided in Supplementary Material.

Each setting uses one frozen operating profile for baseline, detector support, and Doppler-band geometry. The methods companion records the full setting-level whitening recipes and profile details.

Equation (1) defines the baseline ROC score, Equation (2) the fixed detector, Equation (4) the adaptive detector, and Equation (3) the fully whitened detector. Setting-specific whitening estimators, loading values, and training-support policies are fixed before reporting and are listed in the methods companion.

5 Results

Results roadmap. We first report the held-out SIMUS benchmark that determines the default detector configuration. We then show clutter-heavy stress tests that define when whitening helps. Next, we report mechanism ablations on labeled brain simulations and structurally labeled phantom data. Finally, we report bounded real-IQ structural and motion-robustness audits together with replay feasibility.

5.1 SIMUS-Struct benchmark

SIMUS-Struct is a synthetic benchmark motivated by reported fUS operating conditions [8, 9, 34, 47] rather than a generic parameter sweep. We report two prespecified structural settings, SIMUS-Struct-Mobile and SIMUS-Struct-Intraop, with frozen positive, background, and nuisance masks and held-out evaluation on seeds 127–128 after development on seeds 125–126. The positive class is unaliased parenchymal Pf-band flow; the negatives are ordinary background and explicit nuisance structures such as superficial high-velocity or specular components. Throughout this section, $AUC_{\text{main/bg}}$ measures separation from background, $AUC_{\text{main/nuis}}$ measures separation from clinically relevant nuisance, and FPR_{nuis} @ $TPR_{\text{main}} = 0.5$ measures nuisance leakage when half of the target positives are retained.

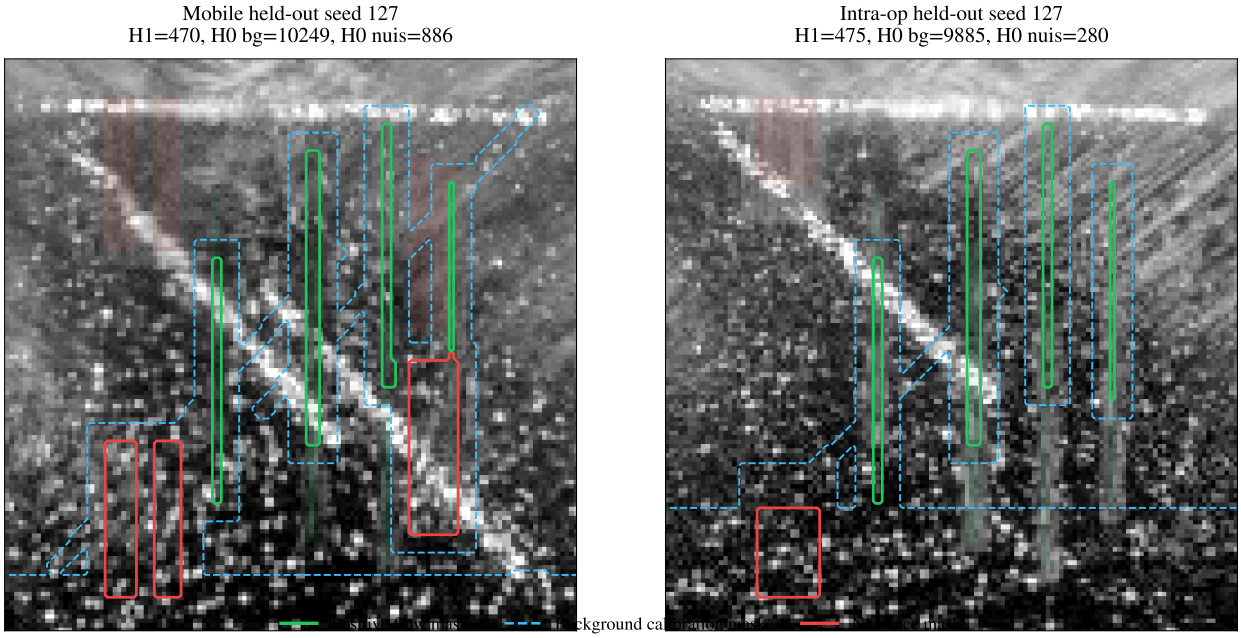


Figure 4: Representative SIMUS-Struct geometry for the SIMUS-Struct-Mobile and SIMUS-Struct-Intraop reference settings. The grayscale underlay is log beamformed power from the simulated IQ cube and is shown only to make the mask geometry visible. Green contours mark the parenchymal-flow positive set H_1, blue dashed contours mark the broad background pool H_0^{bg} used to set the score threshold, and red contours mark the nuisance-negative set H_0^{nuis} used to quantify structured false detections at that same operating point. Table 3 reports the held-out seed results on these frozen mask types across seeds 127–128.

For the functional task, we use the same prespecified settings and a common background-controlled GLM-style readout on simulated task/null runs. We refer to that experiment as the SIMUS functional audit. Figure 4 shows the corresponding structural mask geometry, and Table 3 gives the held-out reference setting that determines the default downstream score.

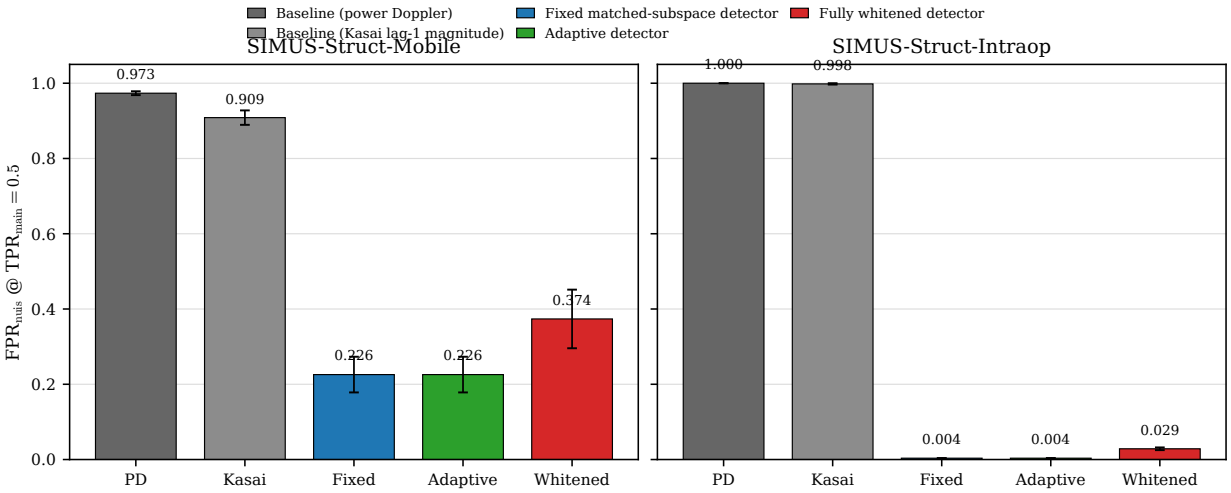


Figure 5: Headline same-residual downstream-head comparison on the held-out SIMUS reference benchmark using the common MC–SVD residual from Table 4. Bars show nuisance false-positive rate at matched main-response recall 0.5, the clearest operating-point summary of the benchmark. Lower bars are better. On the intra-operative setting, the fixed matched-subspace head reduces nuisance FPR from 0.998 for Kasai to 0.004 on the same residual stream; under this frozen guard rule, the adaptive variant matches that result. $\mathrm{AUC}_{\mathrm{main/nuis}}$ values are reported in Table 4.

Mobile

Best conventional baseline: HOSVD → Kasai. Fixed detector chain: Adaptive Global SVD → fixed matched-subspace.

Summary	Best conventional baseline			Fixe	
	$AUC_{main/bg}$	$AUC_{main/nuis}$	FPR_{nuis} @ $TPR_{main} = 0.5$	$AUC_{main/bg}$	AUC
Mean [min,max] over seeds 127–128	0.669 [0.647,0.692]	0.672 [0.624,0.720]	0.286 [0.210,0.361]	0.851 [0.798,0.905]	0.953 [0

Intra-operative parenchymal

Best conventional baseline: HOSVD → Kasai. Fixed detector chain: RPCA → fixed matched-subspace.

Summary	Best conventional baseline			F	
	$AUC_{main/bg}$	$AUC_{main/nuis}$	FPR_{nuis} @ $TPR_{main} = 0.5$	$AUC_{main/bg}$	A
Mean [min,max] over seeds 127–128	0.837 [0.834,0.839]	0.451 [0.431,0.471]	0.621 [0.600,0.643]	0.994 [0.993,0.995]	0.99996

†Exactly zero observed nuisance detections; supported floor 1/886 (mobile) and 1/280 (intra-operative parenchymal).

Table 3: Held-out SIMUS-Struct reference settings. For each evaluated structural setting, we report the best conventional baseline from the development-split baseline sweep and the strongest fixed detector chain chosen on development seeds 125–126, then evaluated unchanged on held-out seeds 127–128. Values are mean [min,max] over the two held-out seeds. $AUC_{main/bg}$ measures separation of target flow from ordinary background, $AUC_{main/nuis}$ measures separation of target flow from explicit nuisance structures, and FPR_{nuis} @ $TPR_{main} = 0.5$ asks how many nuisance detections survive when the detector is tuned to retain half of the target positives.

Table 3 is the held-out reference setting. On the mobile setting, the best conventional baseline among evaluated methods is HOSVD → Kasai, while the strongest fixed detector chain chosen on development seeds 125–126 and evaluated unchanged on held-out seeds 127–128 is Adaptive Global SVD → fixed matched-subspace. On the intra-operative parenchymal setting, the best conventional baseline is again HOSVD → Kasai, while the corresponding fixed detector chain is RPCA → fixed matched-subspace. The headline number is the intra-operative nuisance-tail reduction: at matched main-response recall 0.5, nuisance FPR falls from 0.998 [0.996, 1.000] for Kasai to 0.004 [0.004, 0.004] for the fixed matched-subspace detector. Figure 4 shows the mask geometry used for these audits: thresholds are set on a broad background pool and nuisance leakage is measured on compact structured artifact zones. The fixed-chain rows suppress those nuisance zones completely on all four held-out seed-setting pairs, and Table 3 now reports pooled mean [min,max] summaries over held-out seeds 127 and 128. In Table 3, the 0.000 nuisance rows are literal zero observed nuisance detections on the held-out masks, not rounded near-zero rates. This is the main benchmark result: the band-limited fixed matched-subspace head is the strongest fixed default on held-out structural data, while Table 6 shows where the adaptive and fully whitened variants help in clutter-heavier settings.

To isolate that detector-head claim directly on the main benchmark, Table 4 fixes one held-out MC–SVD residual stream and swaps only the downstream score head. This removes residualizer choice from the comparison. On that fixed residual, PD and Kasai still rank nuisance highly, the fixed matched-subspace head is the strongest same-residual detector, and the fully whitened detector is worse on both held-out settings. The adaptive detector is numerically identical to the fixed head here because its frozen guard-promotion rule does not activate on these cleaner MC–SVD residuals. On this benchmark, the adaptive extension therefore preserves the fixed default when the clutter evidence is weak instead of forcing whitening where it does not help.

Method	Mobile			Intra-operative parenchymal		
	$AUC_{main/bg}$	$AUC_{main/nuis}$	FPR_{nuis} @ $TPR_{main} = 0.5$	$AUC_{main/bg}$	$AUC_{main/nuis}$	FPR_{nuis} @ $TPR_{main} = 0.5$
Baseline (power Doppler)	0.568 [0.562,0.575]	0.131 [0.127,0.134]	0.973 [0.968,0.979]	0.912 [0.909,0.915]	0.022 [0.016,0.028]	1.000 [1.000,1.000]
Baseline (Kasai lag-1 magnitude)	0.500 [0.489,0.512]	0.171 [0.154,0.188]	0.909 [0.889,0.928]	0.792 [0.788,0.795]	0.019 [0.017,0.022]	0.998 [0.996,1.000]
Fixed matched-subspace detector	0.623 [0.587,0.659]	0.713 [0.677,0.748]	0.226 [0.178,0.273]	0.844 [0.831,0.857]	0.982 [0.981,0.983]	0.004 [0.004,0.004]
Adaptive detector	0.623 [0.587,0.659]	0.713 [0.677,0.748]	0.226 [0.178,0.273]	0.844 [0.831,0.857]	0.982 [0.981,0.983]	0.004 [0.004,0.004]
Fully whitened detector	0.569 [0.553,0.585]	0.600 [0.529,0.672]	0.374 [0.296,0.451]	0.819 [0.819,0.820]	0.932 [0.927,0.936]	0.029 [0.025,0.032]

Table 4: Same-residual detector-family ablation on SIMUS-Struct using a common MC–SVD residual. Rows differ only in the downstream score head: power Doppler, Kasai lag-1 magnitude, the fixed flow-band matched-subspace detector without whitening ($R = I$), the adaptive detector that promotes clutter-heavy tiles onto a whitened branch, and the fully whitened detector. Values are means with [min,max] over held-out evaluation seeds 127 and 128 for each setting. Lower nuisance FPR at matched TPR is better.

The key practical question is not whether the method makes vessels look better on average, but whether fewer nuisance detections survive when the detector is tuned to preserve clinically relevant parenchymal flow. That is the practical meaning of FPR_{nuis} @ $TPR_{\text{main}} = 0.5$: it asks how much structured false signal remains at a matched target-flow sensitivity operating point. Because fUS is already being used in intra-operative, bedside, cranial-window, and emerging mobile settings [1, 8, 20–22], that nuisance-leakage tradeoff is tied directly to usability and reader trust rather than to image aesthetics alone.

That per-window operating-point protocol is still an evaluation protocol, not a scanner-time thresholding rule. To address the more practical question directly, we also ran a fixed-calibration transfer check in which the threshold is learned once from separate calibration seeds 125–126 and then applied unchanged to the held-out evaluation seeds 127–128. Table 5 reports that result at nominal $\alpha = 10^{-3}$. On SIMUS-Struct-Mobile, the fixed matched-subspace head preserves nonzero held-out main recall (0.004 [0.002, 0.006]) while lowering realized nuisance FPR from 0.016 [0.015, 0.018] for PD to 0.003 [0.000, 0.006]. On SIMUS-Struct-Intraop, both methods remain near the nominal background target, but only the fixed head suppresses nuisance leakage to 0.000 [0.000, 0.000], whereas PD remains at 0.489 [0.464, 0.514]. The fixed default from the held-out reference setting therefore survives a separate-bank threshold transfer on the benchmark that anchors the paper.

Method	Mobile			Intra-operative parenchymal		
	TPR_{main}	realized FPR_{bg}	realized FPR_{nuis}	TPR_{main}	realized FPR_{bg}	realized FPR_{nuis}
Power Doppler on MC–SVD residual	0.000 [0.000,0.000]	0.002 [0.000,0.003]	0.016 [0.015,0.018]	0.012 [0.006,0.017]	0.001 [0.001,0.001]	0.489 [0.464,0.514]
Fixed matched-subspace head	0.004 [0.002,0.006]	0.002 [0.001,0.003]	0.003 [0.000,0.006]	0.006 [0.004,0.008]	0.001 [0.001,0.001]	0.000 [0.000,0.000]

Table 5: Fixed-calibration transfer on SIMUS-Struct at nominal $\alpha = 10^{-3}$. For each setting, a single threshold is learned once from pooled background negatives on separate calibration seeds 125–126 and then applied unchanged to the held-out evaluation seeds 127–128. Entries report mean [min,max] across the two held-out seeds, so this table is a direct non-retrospective threshold-transfer check rather than the per-window oracle calibration used for ROC reporting elsewhere.

We then turn to harsher nuisance settings on the same benchmark family. These held-out stress rows use the same benchmark infrastructure but push it toward two clinically grounded operating conditions: cardiac-like low-frequency tissue pulsation and progressively shorter slow-time support. The pulsation rows use a fixed 1.30 Hz cardiac-like nuisance term whose amplitude is increased over reference, moderate, and hard settings; this probes the same low-frequency tissue-motion

setting emphasized by Demené *et al.* as critical for SVD clutter filtering and by Baranger *et al.* in neonatal/open-skull Doppler settings [11, 12]. The short-ensemble rows reduce the support from 64 frames to 48 and 32 frames to mimic clinically constrained acquisitions in which shorter windows are used to limit motion corruption or workflow burden [1, 22, 34]. This analysis focuses on the residualizer families that dominated the preliminary reduced-grid search (RPCA and adaptive-global SVD) and then reruns the held-out evaluation at the full reporting tier.

Cardiac-like pulsation

Level / perturbation	Baseline (RPCA→PD)	Fixed head	Adaptive head	Whitened head
Ref.	$0.381^{0.416}_{0.345}$	$0.879^{0.898}_{0.860}$	$0.879^{0.898}_{0.860}$	$0.911^{0.925}_{0.894}$
1.0x cardiac amplitude at 1.30 Hz	$0.743^{0.842}_{0.652}$	$0.034^{0.047}_{0.023}$	$0.034^{0.046}_{0.023}$	$0.017^{0.027}_{0.008}$
Mod.	$0.388^{0.423}_{0.354}$	$0.877^{0.895}_{0.857}$	$0.877^{0.895}_{0.857}$	$0.903^{0.920}_{0.886}$
2.0x cardiac amplitude at 1.30 Hz	$0.668^{0.795}_{0.593}$	$0.049^{0.064}_{0.030}$	$0.049^{0.064}_{0.030}$	$0.030^{0.043}_{0.019}$
Hard	$0.399^{0.433}_{0.365}$	$0.875^{0.893}_{0.854}$	$0.875^{0.894}_{0.854}$	$0.905^{0.921}_{0.890}$
3.0x cardiac amplitude at 1.30 Hz	$0.609^{0.770}_{0.537}$	$0.044^{0.058}_{0.029}$	$0.044^{0.059}_{0.029}$	$0.027^{0.041}_{0.017}$

Short ensemble

Level / perturbation	Baseline (RPCA→PD)	Fixed head	Adaptive head	Whitened head
Ref.	$0.381^{0.415}_{0.345}$	$0.879^{0.897}_{0.861}$	$0.879^{0.898}_{0.859}$	$0.911^{0.926}_{0.895}$
64 frames (42.7 ms)	$0.743^{0.840}_{0.649}$	$0.034^{0.046}_{0.023}$	$0.034^{0.047}_{0.023}$	$0.017^{0.027}_{0.008}$
Mod.	$0.390^{0.423}_{0.356}$	$0.784^{0.807}_{0.758}$	$0.784^{0.808}_{0.759}$	$0.846^{0.868}_{0.824}$
48 frames (32.0 ms)	$0.754^{0.814}_{0.708}$	$0.137^{0.169}_{0.102}$	$0.137^{0.167}_{0.103}$	$0.058^{0.080}_{0.034}$
Hard	$0.426^{0.462}_{0.392}$	$0.772^{0.796}_{0.746}$	$0.772^{0.797}_{0.746}$	$0.716^{0.745}_{0.686}$
32 frames (21.3 ms)	$0.617^{0.727}_{0.536}$	$0.166^{0.194}_{0.139}$	$0.166^{0.193}_{0.139}$	$0.160^{0.207}_{0.130}$

Table 6: Concrete detector-head view of the held-out SIMUS mobile stress regime, fixing the residualizer to RPCA on every row. Entries report $\mathrm{AUC}_{\mathrm{main/nuis}}$ (top) and $\mathrm{FPR}_{\mathrm{nuis}}$@0.5 (bottom), with point estimates shown as $x^{\mathrm{hi}}_{\mathrm{lo}}$ using 95% nonparametric bootstrap intervals over the held-out masked score samples. RPCA is used because RPCA → PD is the strongest conventional baseline among evaluated methods on every reported stress row. This table separates the fixed, adaptive, and fully whitened heads directly instead of reporting only the best family member on each row.

Table 6 is the main held-out stress result. It fixes the residualizer to RPCA and reports the fixed, adaptive, and fully whitened heads separately on the same held-out stress rows. The pattern is coherent without any row-wise family summary: the fully whitened head is strongest on the cardiac-pulsation rows and on the 64-frame and 48-frame short-ensemble rows, while the fixed head remains strongest only on the hardest 32-frame row. On these reduced stress rows the adaptive head is numerically identical to the fixed head, so the family-level advantage is not coming from freely switching among all three modes on every row. A row-wise best-family summary against the strongest conventional baseline is provided in the Supplementary Material. Because this stress setting uses one held-out evaluation seed per axis, the uncertainty notation here is a within-seed bootstrap interval over the held-out masked scores rather than a between-seed interval.

The stress rows complement the reference setting. Together, the stress rows show nontrivial difficulty and the reference rows identify which fixed head should be deployed by default.

Functional same-cube detector-head comparison. Table 7 reports one held-out functional same-cube audit on the same residual stream under an activation-versus-nuisance readout. The

fully whitened detector is directionally positive across most residual families in both settings, but the effect is smaller and less decisive than in the structural benchmark, so we treat it as secondary supporting evidence.

Benchmark setting	Best end-to-end chain	Matched-subspace-positive families	Median Δ selection score [IQR]	Median $\Delta\mathrm{AUC}_{\mathrm{act/nuis}}$ (IQR)
Intra-operative parenchymal	RPCA $\rightarrow$ PD after matched-subspace	5/6	+0.065 [+0.032,+0.091]	+0.026 [+0.014,+0.052]
Mobile	HOSVD $\rightarrow$ PD	5/6	+0.111 [+0.032,+0.142]	+0.059 [+0.011,+0.066]

Table 7: Secondary held-out functional summary on the prespecified SIMUS-Struct settings. This table uses one common background-controlled activation readout for every compared pipeline, so the rows differ only in the upstream residualizer and downstream detector head. "Best end-to-end chain" is the strongest full held-out pipeline on that setting under that common readout. "Matched-subspace-positive families" counts how many of the six prespecified residual families improved when the matched-subspace detector stage replaced the native conventional score on the same residual cube. "PD after matched-subspace" denotes a power-Doppler-style summary map formed after the matched-subspace detector stage. Δ selection score is the held-out pipeline-ranking metric used in this functional benchmark, reported as matched-subspace minus native. $\Delta\mathrm{AUC}_{\mathrm{act/nuis}}$ is the corresponding change in separation between simulated activation and nuisance regions under the same readout. Both deltas are summarized by the median and interquartile range across the six residual families.

Scope and low-FPR operating protocol. Quantitative detection performance is reported on kWave-Stress, Gammex-Struct, and SIMUS-Struct using the strict low-FPR operating protocol of Section 4.1. The ULM and Shin sections add structural, robustness, and transfer audits on open real-IQ data. Conditional execution is a decision-rule variant that uses a flow-proxy mask; leakage ablations and additional diagnostics are reported in Supplementary Material.

5.2 Brain simulation stress tests

The held-out SIMUS-Struct benchmark already determines the default downstream head, so the labeled brain simulations are kept brief in the main preprint. These controlled kWave-Stress audits ask a narrower mechanism question on the same MC–SVD residual cube: does the gain come from band selectivity, from whitening alone, or from their combination? The answer is that whitening alone is not sufficient in these stress tests. On fixed residuals, whitened total power collapses at the reported strict-tail operating points, whereas the fixed band-limited matched-subspace score remains strong; companion tables also show that detector-swap gains do not automatically persist on more aggressive residualizers and that the fully whitened variant remains stable under modest covariance-training trim changes. Taken together, these labeled stress tests explain why the held-out SIMUS benchmark favors the fixed head and why whitening becomes useful only in narrower clutter-heavy settings. The full detector-component tables, swap audit, and labeled-brain ROC figures are provided in the methods companion.

5.3 Gammex phantom: structural ROC and within-ensemble motion ladders

Dataset and acquisition parameters. We use the open-access Doppler Twinkling Artifact database [42], which provides complex beamformer output from phantom acquisitions on a clinical scanner. The *Gammex flow phantom* provides a simple vessel-like geometry that lets us derive independent structure-based flow/background labels from B-mode anatomy alone. Each cine frame contains a contemporaneous B-mode image and a short-ensemble color-flow (CFM) slow-time block with $N \in \{5, 9, 17\}$ pulses. Leonov *et al.* explicitly report PRF for some acquisition families, including the transverse linear-probe Gammex acquisition, but also note that PRF information is not consistently preserved across the database. We therefore fix PRF once per sequence family, using the source-paper value where it is explicitly documented and otherwise recording a sequence-level override in bundle metadata. Band-discretization sanity checks flag invalid Pf/Pa discretizations explicitly. Decoding/QC details are provided in Supplementary Material (PRF provenance and decoder sanity); reproduction commands and manifests are provided in Supplementary Material. Additional artifact-heavy phantom sanity checks are kept in the methods companion.

B-mode-only structural tube masks (independent H_1/H_0). For the Gammex flow phantom sequences we derive evaluation masks from B-mode structure only (no Doppler or matched-subspace outputs). Using acquisition geometry metadata that specifies the CFM region-of-interest (ROI) within the B-mode grid (beam and sample offsets and any beam subsampling), we map the B-mode ROI onto the CFM grid and segment the tube lumen on a median log-envelope reference image. This yields (i) a conservative lumen (flow) mask, eroded to avoid wall pixels and (ii) a background mask that excludes a guard band around the lumen and bright reflectors. Mask provenance and QC metrics are recorded with each experiment; Figure 6 shows representative overlays. These structural labels support a defensible phantom ROC.

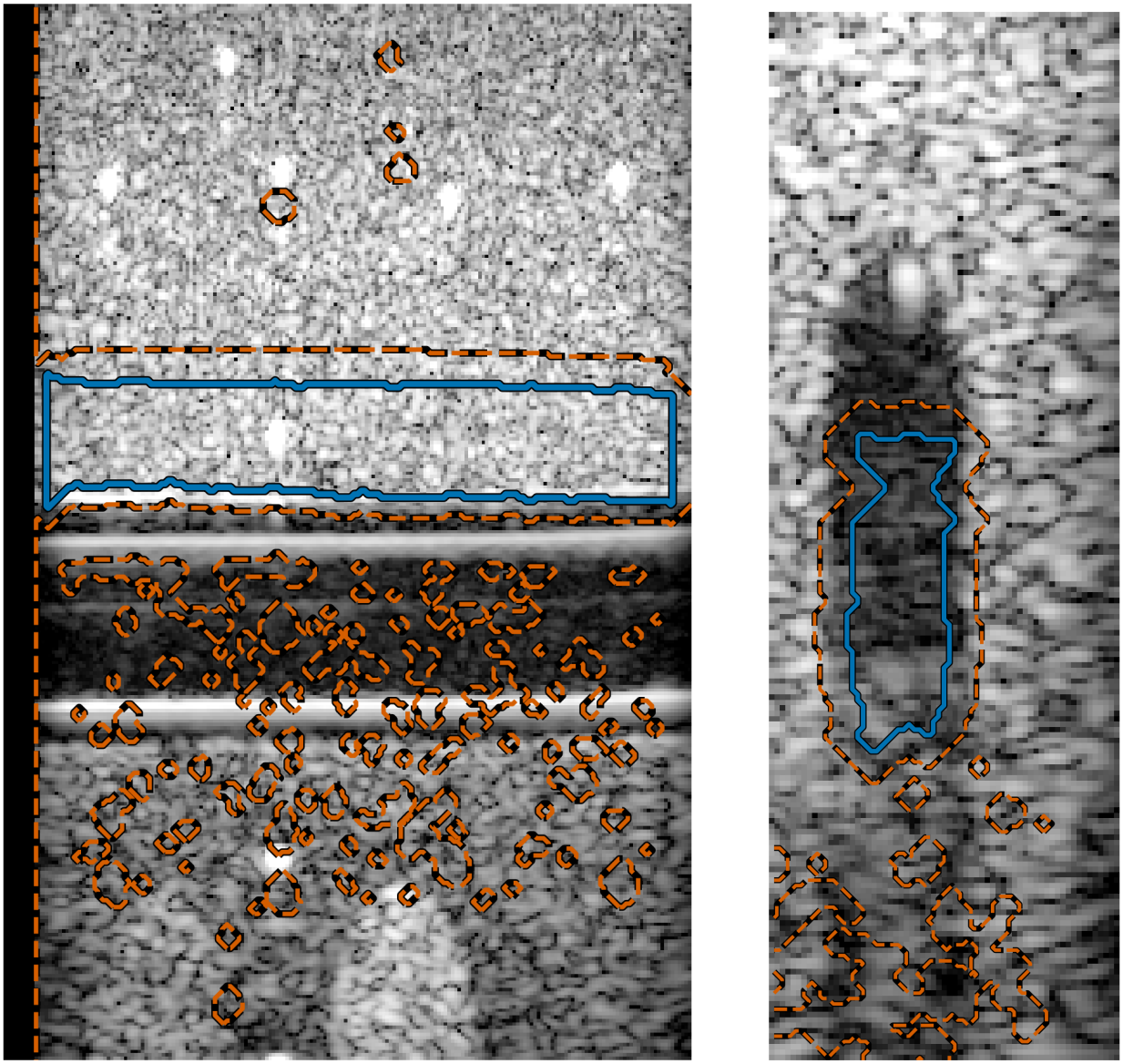


Figure 6: Gammex flow phantom structural evaluation masks derived from B-mode only. Left: along-linear view; right: across-linear view. The lumen mask is intentionally eroded to avoid wall/partial-volume pixels, and the background mask excludes a guard band around the lumen and bright reflectors. Images show the CFM-aligned B-mode ROI with contour overlays of the tube lumen (flow) mask (blue) and background mask (orange, dashed) used for structural-label ROC reporting. Reproducibility details are provided in Supplementary Material.

Frozen short-ensemble Doppler profile and baseline fairness scores. For the linear-probe Gammex sequences used here ($N = 17$ shots; PRF values listed in Supplementary Material), we use a prespecified short-ensemble operating profile: baseline SVD band-pass keep range $[2, 17]$, detector aperture $L_t = 16$, tiles 8×8, a Tyler-type robust covariance estimator, and diagonal

loading $\lambda = 0.07$. To keep the profile auditable while avoiding sample-support limitations on small ROIs and excessive overlap/mixing on large ROIs, we use a deterministic geometry-only stride policy: choose the largest tile stride ≤ 6 such that the implied tile count is ≥ 500. On the two Gammex views this yields stride 6 (along; large ROI) and stride 4 (across; small ROI). Band definitions are set to $P_f = [150, 450]$ Hz and $P_a = [700, 1200]$ Hz. For baseline context on short ensembles, we also report conventional comparators derived from the baseline-filtered cube (power Doppler, log power Doppler, and Kasai lag-1 power). Reproducibility details are provided in Supplementary Material.

Structural-label operating points (Gammex). The same strict-tail geometry of Figure 3 appears in the short-ensemble Gammex phantom, where residual twinkling structure dominates the upper background tail. We therefore pool pixel scores across cine frames and report low-FPR operating points using right-tail thresholding on pooled background pixels, then evaluate TPR on pooled lumen pixels. In these short-ensemble Doppler acquisitions, the background set includes strong artifact pixels whose right-tail scores dominate conventional baselines; consequently, low-FPR thresholds at $\alpha \leq 10^{-3}$ can lie above essentially all lumen pixels, yielding numerically zero baseline TPR at the reported operating points. Figure 8 shows one representative pooled baseline comparator. To avoid treating pooled pixels as independent samples, we report 95% confidence intervals via a frame-level bootstrap: we resample cine frames as the unit of variation while holding the pooled threshold fixed. Table 8 summarizes a broader same-residual score comparison for the two Gammex views under those same settings: power Doppler and Kasai on the baseline residual, plus three detector-family score variants evaluated on that identical residual cube (whitened power, the fixed matched-subspace detector, and the fully whitened matched-subspace detector). For compactness we omit log-PD from the main table because it is monotone-equivalent to PD here and numerically identical at the reported operating points; the full pooled summaries remain available in Supplementary Material. The practical relevance is already visible at a conventional operating point. On the representative Figure 7 frame at matched background FPR 10^{-2}, the baseline thresholded map contains 884 suprathreshold pixels and none lie inside the structural lumen mask, whereas the matched-subspace score recovers 586 lumen pixels while holding the same background allowance. Table 8 now reports that relaxed $\alpha = 10^{-2}$ point alongside the stricter tail operating points, so the phantom result can be read both as a practitioner-facing map-quality summary and as a stress test. The same-residual phantom comparison shows that the baseline residual still contains recoverable lumen signal, because changing only the downstream score restores nonzero sensitivity. In this short-ensemble setting, local whitening is the dominant ingredient, and total whitened slow-time power is stronger than the matched-subspace ratio on both views, especially across-view. Gammex therefore supports localized covariance-aware downstream scoring, but not the claim that the matched-subspace ratio is the preferred score in every phantom setting. The view asymmetry still matters: the across view provides the clearest structural-label phantom evidence, whereas the along view remains a harsher artifact-tail stress test under this short-ensemble profile. A small untuned sanity audit in Supplementary Material gives similar values: on the first 20 cine frames, applying the matched-subspace detector directly to raw registered IQ changes TPR@10^{-3} only from 3.48×10^{-4} to 2.61×10^{-4} on the along view and from 0.202 to 0.205 on the across view; perturbing the frozen along-view PRF from 2000 to 3000 Hz leaves TPR@10^{-3} unchanged at 3.48×10^{-4} in that limited audit.

	Along (linear17 @ PRF 2500)				Across (linear17 @ PRF 2500)			
Method / score	TPR@$\alpha = 10^{-2}$	TPR@$\alpha = 10^{-4}$	TPR@$\alpha = 3\times10^{-4}$	TPR@$\alpha = 10^{-3}$	TPR@$\alpha = 10^{-2}$	TPR@$\alpha = 10^{-4}$	TPR@$\alpha = 3\times10^{-4}$	TPR@$\alpha = 10^{-3}$
Baseline (power Doppler)	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]
Baseline (Kasai lag-1 power)	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]
Detector ablation (whitened power)	0.4116 [0.4078,0.4155]	0.0012 [0.0011,0.0014]	0.0119 [0.0114,0.0124]	0.0532 [0.0521,0.0543]	0.7234 [0.7206,0.7262]	0.4820 [0.4780,0.4863]	0.5155 [0.5115,0.5191]	0.5813 [0.5783,0.5843]
Detector ablation (unwhitened ratio)	0.00031 [0.00022,0.00041]	0.00000 [0.00000,0.00001]	0.00000 [0.00000,0.00001]	0.00001 [0.00000,0.00002]	0.1095 [0.1040,0.1148]	0.0574 [0.0533,0.0611]	0.0602 [0.0562,0.0643]	0.0653 [0.0610,0.0695]
Fully whitened matched-subspace detector	0.0942 [0.0926,0.0958]	0.00000 [0.00000,0.00000]	0.00003 [0.00000,0.00007]	0.00031 [0.00020,0.00043]	0.6891 [0.6846,0.6939]	0.0491 [0.0469,0.0513]	0.0998 [0.0970,0.1027]	0.2237 [0.2197,0.2276]

Table 8: Gammex flow phantom structural-label operating points using B-mode-derived structural masks. All scores are right-tail (higher indicates more flow evidence). This is a same-residual detector-head comparison: every row is evaluated on the identical baseline SVD band-pass residual cube, changing only the detector head. The baseline rows are power Doppler and Kasai lag-1 power on that residual; the detector ablations replace the fully whitened matched-subspace detector with either total whitened slow-time power (no Doppler band partition) or the same flow-band matched-subspace ratio without covariance whitening ($R = I$). The table includes $\alpha = 10^{-2}$ as a practitioner-facing relaxed operating point together with the stricter $\alpha \leq 10^{-3}$ tail regime used for stress testing. Thresholds are chosen per method on pooled background pixels to match each target FPR and evaluated on pooled lumen pixels. Brackets show 95% frame-bootstrap CIs ($n = 2000$ resamples over cine frames) at the fixed pooled threshold. Pooled background counts are $n_{\text{bg}} = 3.85 \times 10^6$ for the along view ($fpr_{\min} \approx 2.6 \times 10^{-7}$) and $n_{\text{bg}} = 1.97 \times 10^6$ for the across view ($fpr_{\min} \approx 5.1 \times 10^{-7}$). Reproducibility details are provided in Supplementary Material.

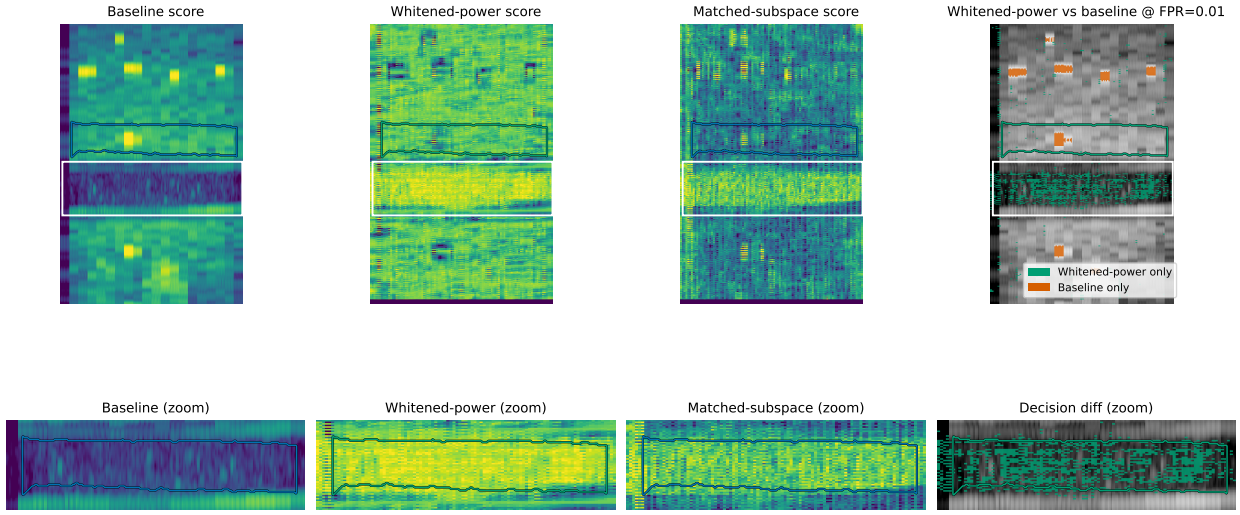


Figure 7: Leading structural-fidelity visual on a structurally labeled Gammex flow phantom acquisition from the open Doppler Twinkling Artifact database (labels derived from B-mode-only tube segmentation). Top row: baseline power Doppler image S_{base} (SVD band-pass clutter suppression followed by power Doppler), the whitened-power map, the matched-subspace map, and a matched-FPR decision-difference panel at a fixed background false-positive rate (10^{-2}; chosen here for visualization), showing pixels detected only by whitened power versus only by the baseline. Bottom row: zoomed region of interest. The tube contour is overlaid for reference. The qualitative ordering matches Table 8: whitened power is the strongest phantom score in this short-ensemble setting, while the matched-subspace panel remains included as the detector-core reference.

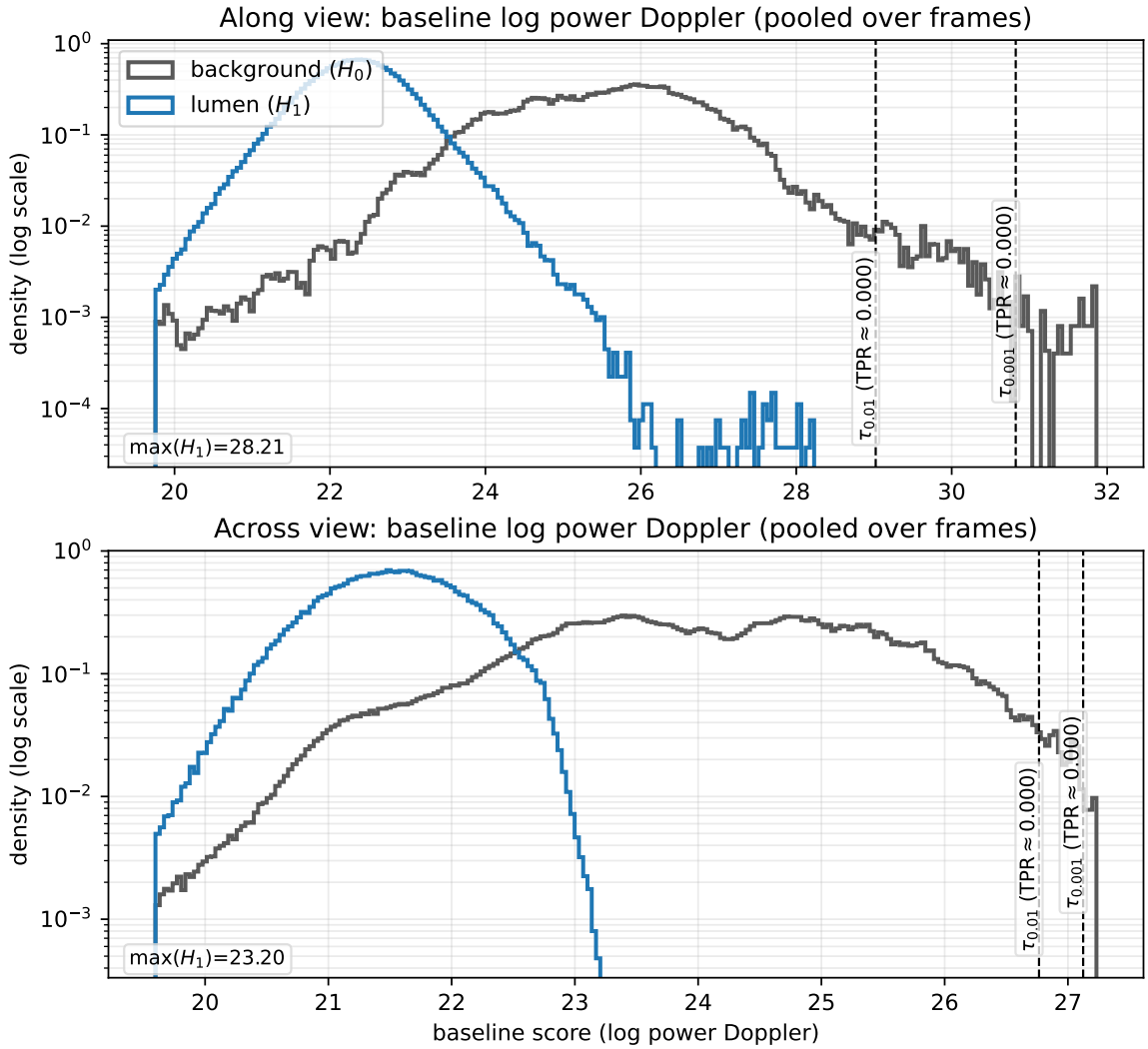


Figure 8: Why conventional baselines can have zero TPR at ultra-low FPR on Gammex short-ensemble data. The plot shows pooled score distributions for the baseline log power Doppler comparator on lumen (H_1; structural tube mask) versus background (H_0; structural background mask), for both along and across views. Vertical lines indicate background-calibrated right-tail thresholds τ_α at $\alpha \in \{10^{-2}, 10^{-3}\}$; $\alpha = 10^{-2}$ is shown here for context because the reported operating points are $\alpha \leq 10^{-3}$. On both views the background tail threshold lies above $\max(H_1)$, so $\mathrm{TPR}(\alpha) = \mathbb{P}\{S \geq \tau_\alpha \mid H_1\}$ is numerically zero at these tail operating points.

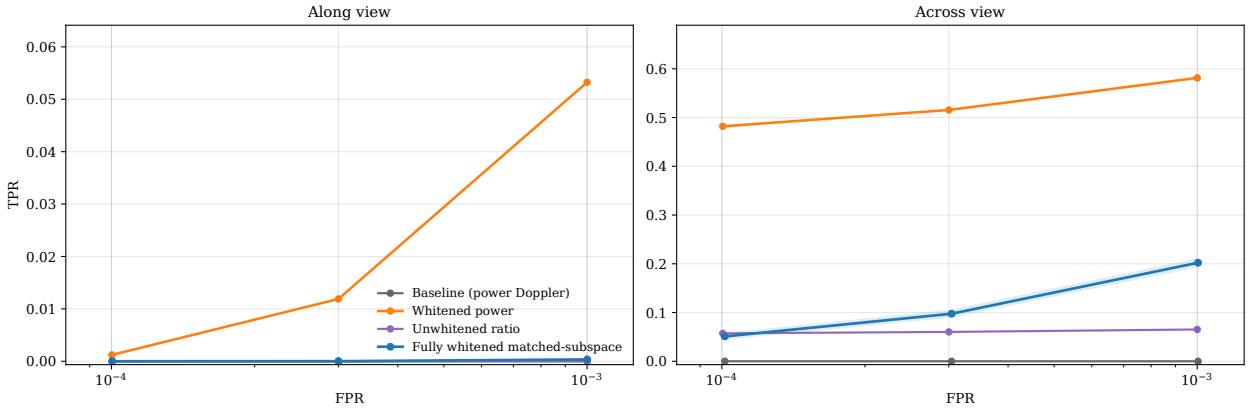


Figure 9: Gammex flow phantom same-residual score comparison in the strict low-FPR region used for the main phantom table. Curves connect the three reported operating points for baseline power Doppler, whitened power, the fixed matched-subspace detector, and the fully whitened matched-subspace detector. The two panels use panel-specific TPR ranges so that the operating-point differences are readable.

Additional Gammex sanity checks are provided in the Supplementary Material.

Within-ensemble motion ladders. A labeled within-ensemble motion ladder on the along-view Gammex cine is included in the methods companion. It is directionally consistent with the main phantom result and is useful as a robustness check, but the central phantom claim in the main preprint is the same-residual structural ROC comparison above.

5.4 ULM real-IQ audits

Dataset and loader. We use a real kHz IQ dataset released for ultrasound localization microscopy benchmarking [37, 38] and distributed as HDF5 blocks with accompanying acquisition metadata. Each block contains a dataset `iq` of shape $(2, H, W, T) = (2, 256, 180, 800)$ float32 representing I/Q over $T = 800$ slow-time frames. We load blocks on demand and return complex IQ as a cube of shape (T, H, W) (reproduction details in Supplementary Material).

One-time label-free baseline selection. For ULM we select a single MC–SVD energy-fraction cutoff by a one-time label-free sweep on blocks 1–3 over frames 0:128, and then freeze it for all subsequent ULM motion experiments. The sweep uses baseline-only runs and relies on label-free integrity and stability criteria (no task labels); it yields a frozen energy fraction $e = 0.975$ (full sweep details in Supplementary Material). The later PALA-backed structural audit and block-split threshold-transfer analysis use the same open dataset and therefore overlap in source blocks, but this one-time sweep used no localization-derived structural labels and fixed only the baseline MC–SVD energy fraction before those analyses were run.

5.4.1 Localization-derived structural audit

ULM-CoreShell. To obtain a more anatomical structural readout on the same open real-IQ data, we use the published PALA rat-brain ULM example as an external vascular reference derived from

microbubble localization microscopy rather than from the Doppler scores being evaluated [37, 38]. We first reproduce the corresponding short-horizon PALA PowDop rendering from the Zenodo IQ blocks using the published odd-block selection and SVD cutoff, register that published PowDop crop back to the Zenodo IQ field of view, and then downsample the published super-resolution MatOut vascular rendering onto the short-window evaluation lattice. From that registered localization map we derive a conservative vessel-core versus perivascular-shell label pair: the vessel mask is a peak-based core mask, and the background mask is the surrounding shell, so the endpoint asks whether a short-window detector concentrates score on the vascular core rather than the immediately adjacent parenchyma. That one-time PALA-to-IQ registration is part of the label construction only. For the short-window detector itself we keep frame-to-frame registration disabled in the frozen audit, because re-enabling it changes the structural ROC only below the fourth decimal place while adding substantial latency.

On this endpoint, Table 1 reports PD, Kasai, the fixed head, the adaptive head, and the fully whitened variant on the same residual windows and the same localization-derived vessel-core versus shell masks. The fixed and adaptive heads yield only modest pooled-ROC gains over PD and Kasai, and adaptive is nearly indistinguishable from fixed under this profile. The fully whitened variant is the only head that improves pooled AUC, strict-tail sensitivity, and shell FPR at matched 70% vessel-core recall: AUC 0.530 [0.524, 0.537], TPR@10^{-3} 0.004 [0.003, 0.006], and shell FPR 0.676 [0.665, 0.687], compared with AUC 0.514/0.514, TPR@10^{-3} 0.000/0.000, and shell FPR 0.703/0.702 for the fixed/adaptive heads and AUC 0.504/0.503, TPR@10^{-3} 0.000/0.000, and shell FPR 0.698/0.701 for PD/Kasai.

The pooled advantage is not driven by one favorable block. Relative to the fixed head, the fully whitened variant exceeds AUC on 47/70 audited windows and all 10/10 blocks, lowers shell FPR at matched 70% core recall on 47/70 windows and 9/10 blocks, and yields nonzero TPR@10^{-3} in 45/70 windows versus 10/70 for the fixed and adaptive heads (Table 9). The same pattern holds against the adaptive head: 46/70 AUC windows, 47/70 shell-FPR windows, and 10/10 blocks. Against baseline power Doppler, the fully whitened variant also wins AUC on all 10 audited blocks, and an exact paired sign-flip test on the block-mean AUC difference gives $p = 0.00195$. Figure 1 gives the visual counterpart to this result by showing a representative same-window comparison between baseline power Doppler and the fully whitened variant. On an RTX 4080 SUPER, the same 64-frame no-registration configuration runs at 41.3 ms steady-state total for a 64 ms acquisition budget under the reported replay conditions. The labels come from a published localization-derived vascular rendering on the same acquisition rather than from an independent modality or task-based ground truth, so this remains a structural rather than functional result.

To reduce overlap between localization-derived label construction and scored windows, we also ran a stricter odd-block-reference / even-block-evaluation variant under the same frozen 64-frame profile. The family ordering remained directionally consistent: the fully whitened variant still led on pooled AUC and strict-tail sensitivity, though with smaller margins than in the main same-acquisition audit. This stricter split reduces, but does not eliminate, the same-acquisition overlap limitation, and the summary is provided in the Supplementary Material. A supplementary falsification control points in the same direction: support-matched randomization of the vessel-core and shell masks collapses the structural audit toward chance, while a large cyclic translation destroys the fully whitened ordering rather than reproducing it.

Comparison	AUC wins (windows / blocks)	shell-FPR wins @70% core recall (windows / blocks)	nonzero TPR@10^{-3} windows
Fully whitened variant vs fixed head	47/70 / 10/10	47/70 / 9/10	45/70 vs 10/70
Fully whitened variant vs adaptive head	46/70 / 10/10	47/70 / 10/10	45/70 vs 10/70

Table 9: Window-level consistency summary for the PALA-backed ULM structural audit. Each row compares the prespecified fully whitened variant against another detector head on the same frozen 64-frame MC–SVD residual cube and the same localization-derived vessel-core versus shell labels. The window/block entries report how often the fully whitened variant has higher vessel-core AUC or lower shell false-positive rate at matched 70% vessel-core recall.

A second transfer analysis uses the same structural endpoint but removes per-window retuning. We split the 10 audited blocks into two complementary 5-block calibration/evaluation halves, learn a single shell-calibrated threshold on one half, and apply it unchanged to the other half at nominal $\alpha = 10^{-3}$. Table 10 shows the result. Under that fixed threshold transfer, PD and Kasai collapse to zero held-out vessel-core recall, whereas the fully whitened matched-subspace variant retains a small but nonzero held-out core TPR of 0.002 [0.000, 0.004] at a realized shell FPR of 0.001 [0.000, 0.002]. More concretely, it is the only score with any positive held-out core detections at all, firing on 6 of the 20 held-out windows versus 0 of 20 for PD and Kasai. The effect is still modest, but it shows that the structural claim does not depend entirely on per-window threshold readjustment.

Score	core TPR	realized shell FPR
Baseline (power Doppler)	0.000 [0.000,0.000]	0.001 [0.000,0.002]
Baseline (Kasai lag-1 magnitude)	0.000 [0.000,0.000]	0.001 [0.000,0.002]
Whitened matched-subspace variant	0.002 [0.000,0.004]	0.001 [0.000,0.002]

Table 10: Fixed-calibration transfer on the PALA-backed ULM structural audit at nominal $\alpha = 10^{-3}$. A threshold is learned once from pooled perivascular-shell negatives on one 5-block calibration bank and then applied unchanged to the complementary 5 held-out blocks; the reported values pool the two complementary block splits and show window-level means with 95% bootstrap CIs over held-out windows. This is therefore a genuine non-retrospective threshold-transfer check on real in-vivo IQ rather than per-block retuning.

5.4.2 Motion robustness audit

Companion motion audit. A companion-only motion-injection audit on ULM-CoreShell and Shin-ProxyMotion provides secondary robustness evidence under controlled residual-motion stress. In those audits, matched-subspace score maps are more stable than the baseline maps, but these results are robustness corroboration rather than the structural efficacy claim that anchors the main real-IQ section.

Shin-ProxyMotion. To avoid making the motion-robustness story depend on a single open dataset, we also summarize the strongest existing Shin rat-brain motion-stress result from the supplement [35,36]. Under a prespecified low-frequency operating setting and a controlled residual-motion stress test, the matched-subspace map is more stable than the MC–SVD baseline map across all three audited windows, with self-correlation improving from roughly 0.18 to 0.36–0.45 and a nonzero flow-proxy hit rate appearing only for the matched-subspace statistic. A separate fixed-threshold check on the same proxy masks remains negative: when one background-proxy threshold is learned on five clips and applied unchanged to the complementary five clips at nominal $\alpha = 10^{-3}$,

PD, Kasai, and the fixed matched-subspace detector all fall to zero held-out flow-proxy hit rate at realized background-proxy FPR near target. Full Shin tables are provided in the methods companion.

The real-data results span four roles: Gammex-Struct provides structurally labeled phantom evidence, ULM-CoreShell adds a structural check using a localization-derived vascular reference plus block-split threshold transfer, ULM-Motion provides open real-IQ robustness evidence, and Shin-ProxyMotion provides a second open real-IQ corroboration under controlled residual-motion stress plus a negative cross-clip proxy-transfer check. A compact same-cube score audit across those real-IQ bundles is kept in the Supplementary Material, together with additional label-free audits such as whole-brain PD-only retrospectives, Shin contract summaries, and extended motion variants.

5.5 Computational cost and replay feasibility

We also report representative runtimes for the final detector family evaluated in this manuscript rather than for a whitening-heavy research configuration. In this evaluation, the labeled-brain stress tests use the fixed matched-subspace default with conditional execution, the Shin real-IQ evaluation uses the adaptive variant, and Gammex uses the short-ensemble raw-baseline plus fixed matched-subspace head. Conditional execution changes the decision rule by deciding which tiles receive the adaptive detector, so it is reported as part of the detector configuration rather than as a free acceleration of the same method. Unless otherwise stated, conditional-execution reporting uses the same leakage-controlled, disjoint-window proxy discipline as the detection results.

Latency is measured by replaying multiple windows or frames in a single GPU process and reporting *steady-state* means over windows/frames 2..N. The first replay window/frame is excluded because it includes one-time startup overheads. We time the baseline and detector stages separately, excluding disk I/O, and "Total" reports baseline plus detector wall time. "Budget" denotes the acquired slow-time duration of the corresponding ensemble: $64/1500 \approx 42.7$ ms for the brain windows, $128/1000 = 128$ ms for Shin, and $17/2500 = 6.8$ ms for the Gammex cines. The reported real-time factor (RTF) is therefore Total/Budget.

Scenario	Baseline (s)	Detector (s)	Total (s)	Budget (s)	RTF
Open-skull surface-artifact stress test (64-frame window)	0.024	0.012	0.037	0.043	0.86
Alias-dominant positive control (64-frame window)	0.025	0.012	0.038	0.043	0.88
Structured-clutter leakage stress test (64-frame window)	0.026	0.012	0.037	0.043	0.87
Shin rat-brain real-IQ evaluation (128-frame window)	0.006	0.102	0.108	0.128	0.84
Gammex flow phantom, along view (17-shot cine frame)	0.000	0.006	0.006	0.007	0.88
Gammex flow phantom, across view (17-shot cine frame)	0.000	0.004	0.004	0.007	0.64

Table 11: Steady-state latency summary on an NVIDIA GeForce RTX 4080 SUPER 16GB for the final detector family evaluated in this manuscript. Times are means over windows/frames 2..N, exclude disk I/O, and compare baseline and detector stages under the reported detector configuration rather than under an earlier whitening-heavy timing setup.

Table 11 shows that the replay picture is substantially better in the final reported configurations than in a dense fully whitened setup. All three labeled-brain stress tests run below the full acquisition-PRF budget (RTF $\approx$ 0.86–0.88), the Shin rat-brain real-IQ evaluation also remains below budget (RTF $\approx$ 0.84), and both Gammex views run below the 17-shot cine-frame budget (RTF $\approx$ 0.88 along-view and $\approx$ 0.64 across-view).[1] These timings support workstation replay feasibility for the reported configurations, but not scanner-integrated or cross-session deployment.

A leakage ablation confirms that conditional execution can inflate apparent performance under same-window proxy gating, while leakage-controlled gating yields a small TPR decrease (order 10^{-3}) at the target FPRs. The full open-skull leakage table with bootstrap intervals is provided in the Supplementary Material.

6 Discussion

What the detector family adds

Post-beamformed fUS benefits from a better detection statistic, not only from a better clutter filter. Ordinary clutter suppression can still leave structured right-tail false detections, and the central result of this paper is that changing only the final scoring function can materially change artifact suppression on the same clutter-filtered residual cube. The mechanism depends on setting: matched-subspace geometry is the main driver on the labeled-brain stress tests and the held-out SIMUS benchmark, whereas local slow-time whitening becomes more important in short-ensemble or structure-concentration settings. Gammex-Struct is the clearest counterexample to a universal matched-subspace claim: on the same residual phantom data, whitened power is stronger than the matched-subspace ratio on both views. These results define a detector family with a clear

[1] Latency is reported from steady-state replay (windows/frames 2..N) and excludes startup overhead and disk I/O. Table 11 reports the final detector family used in the manuscript rather than a single worst-case fully whitened configuration applied everywhere.

fixed default and narrower whitened variants rather than one statistic or one mechanism that wins everywhere.

For a practitioner, the immediate implication is that the same SVD residual already used in existing pipelines can yield a more trustworthy Doppler map by changing only the final scoring step. The fixed head requires no additional training data and no retuning of the upstream clutter filter. On the held-out benchmark, it eliminates all nuisance detections at matched 50% target-flow recall on the same residual where power Doppler passes 97–100% of nuisance. That is not a marginal image-quality change; it is the difference between a map that can be thresholded with confidence and one that still requires visual inspection to separate artifact from signal. Whether that difference transfers fully to real *in vivo* task data remains the central open question.

The comparator set is intentionally clutter-filter-heavy rather than learning-heavy because the question here is the downstream detection statistic on a fixed clutter-filtered residual. Learned clutter filters remain important related work, but comparing against them fairly would require a separate supervised training protocol and is left for future work.

The strict-tail operating points in the labeled experiments use known negative sets, so they are an evaluation protocol rather than a scanner-time thresholding rule. The transfer checks show some non-retrospective signal on SIMUS-Struct and ULM-CoreShell, while Shin remains negative, so a practical thresholding workflow on raw *in vivo* IQ remains unestablished.

Real-data scope and clinical interpretation

Most public *in vivo* fUS releases do not preserve the raw high-PRF complex IQ and metadata needed for independent detector evaluation after beamforming. For that reason, the strongest efficacy evidence in this paper still comes from labeled simulations and structurally labeled phantom data, while the open real-IQ sections are narrower robustness and structural audits. The ULM-CoreShell result is directionally positive but modest, and only the fully whitened variant shows a clear advantage there. The real-data contribution is therefore improved vascular and artifact discrimination at the detector stage, not biological inference or real-data functional task validation.

A stricter disjoint-source odd/even block audit remained directionally consistent but weaker, and a support-matched permutation control collapsed the structural audit toward chance. Together these checks reduce the simplest same-acquisition leakage objections, but they do not turn the result into independent multimodal or cross-session validation.

The simulation-to-reality gap should calibrate expectations: the held-out synthetic benchmark determines the default detector choice much more cleanly than the current real-IQ audits. What is still missing is a standardized open benchmark with repeated sessions and independent anatomical or task anchors for post-beamformed detector evaluation [8, 22–25].

Recommended detector variant by setting

Figure 10 summarizes the practical deployment rule supported by the current evidence. The fixed head remains the default, while the fully whitened variant is reserved for repeated guard-dominant windows with nonzero promotion and sufficient latency margin; the current adaptive telemetry is useful for label-free monitoring, but not yet a validated universal selector.

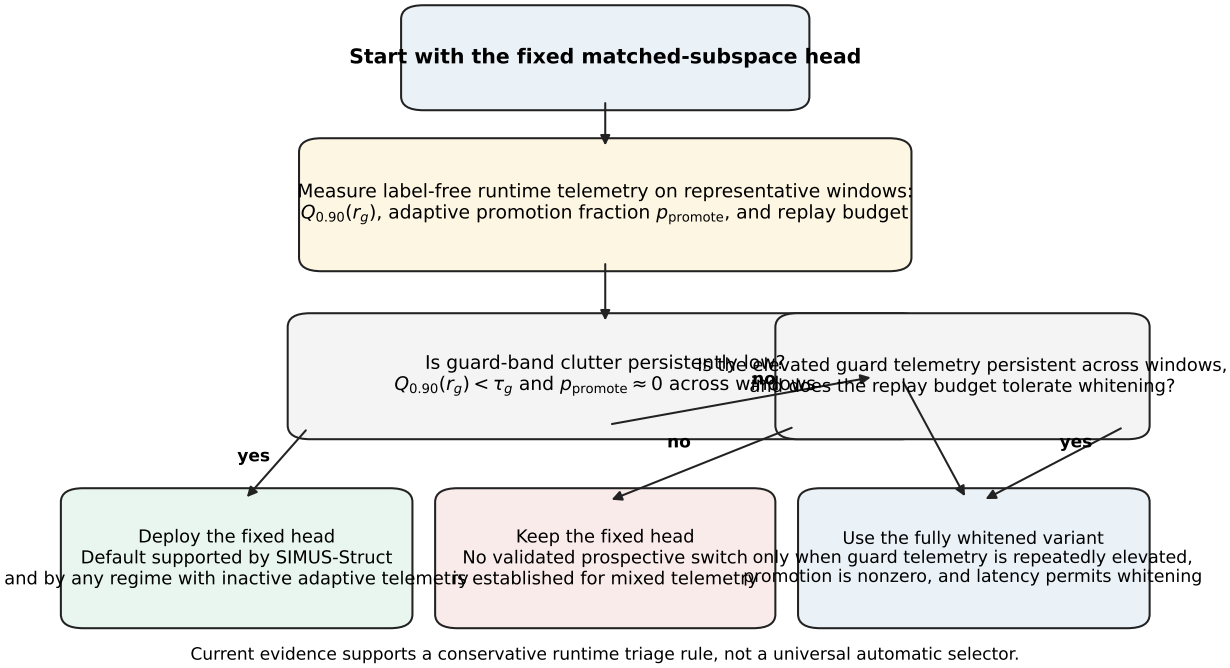


Figure 10: Conservative label-free deployment rule for choosing between the fixed and fully whitened detector variants. The runtime telemetry is already produced by the adaptive branch: r_g is the tile-level guard-band energy fraction, $Q_{0.90}(r_g)$ is its upper-tail summary across a representative window, and p_{promote} is the fraction of tiles that would be promoted onto the whitened branch by the frozen adaptive rule. The flowchart encodes the present practical conclusion: fixed is the default, fully whitened is reserved for repeated guard-dominant windows with nonzero promotion and sufficient latency budget, and mixed telemetry is not enough to justify an automatic switch. This is a conservative deployment heuristic from the current evidence, not a validated universal selector.

7 Conclusion and future work

We presented a localized matched-subspace detector framework for post-beamformed fUS and ultrafast Doppler data, centered on a band-limited fixed head and extended by an adaptive whitening switch plus a fully whitened variant. Across the evaluated labeled simulations, SIMUS-Struct, and one structurally labeled phantom setting, this detector family often improved low-FPR nuisance control relative to PD/Kasai-style downstream scores, but the preferred downstream head depended on setting. Across those tiers, the fixed band-limited head remains the structural default, while the adaptive and fully whitened variants help in selected clutter-heavy settings rather than serving as universal replacements. The phantom result should be read accordingly: it supports localized covariance-aware downstream scoring, but not a universal matched-subspace mechanism.

On one open real-IQ rat-brain dataset, the same-acquisition ULM structural audit shows only modest gains for the fixed and adaptive heads, while the fully whitened variant is the only head that clearly improves pooled AUC, strict-tail sensitivity, and shell suppression on the vessel-core

versus perivascular-shell endpoint. A stricter disjoint-source odd/even block variant preserved that ordering with attenuated margins, so the real-IQ structural signal is not wholly dependent on block overlap inside the same acquisition. Fixed-threshold transfer remains only partly established: SIMUS-Struct and the ULM structural audit retain non-retrospective signal, while the Shin proxy transfer stays negative. This is a detector-stage contribution for post-beamformed fUS pipelines, not completed real-data functional validation. The main next external test is prospective labeled *in vivo* validation on raw high-PRF IQ with independent anatomical or task anchors and a practical non-retrospective thresholding workflow.

More broadly, freezing the clutter-suppressed cube and comparing only the downstream score provides a general protocol for benchmarking post-beamformed fUS detection methods and could serve as a standardized evaluation layer as the field moves toward open reproducible pipelines.

Data and code availability

All datasets used in this work are either publicly available (as cited) or derived deterministically from those sources. All code used to generate the reported tables and figures is versioned; the exact commit hash and release information are recorded in the reproducibility manifest (Supplementary Material). A minimal CUDA-enabled software environment and container recipe are provided in the repository. End-to-end regeneration commands for the reported figures and tables are collected in the repository and summarized in the reproducibility appendix.

Code repository. The reference implementation and reproduction scripts are available at `https://github.com/ArthurShune/fus-detectors`. The canonical commit hash and release information are recorded in the reproducibility manifest (Supplementary Material).

References

[1] Imbault, M., et al. Intraoperative functional ultrasound imaging of human brain activity. *Scientific Reports* 7, 7304 (2017).

[2] Macé, E., Montaldo, G., Pernot, M., Cohen, I., and Tanter, M. Functional ultrasound imaging of the brain. *NeuroImage* 54(4), 2720–2731 (2011).

[3] Urban, A., et al. In vivo whole brain microvascular imaging in mice using ultrafast ultrasound. *EBioMedicine* 76, 103861 (2022).

[4] Alexandrov, A. V., et al. Velocity measurements in the middle cerebral arteries of healthy adults by transcranial Doppler. *Journal of Neuroimaging* 1, 166–172 (1991).

[5] Alexandrov, A. V., et al. Optimal values of flow velocity on transcranial Doppler in patients with intracranial arterial stenosis. *Stroke* 33, 1757–1761 (2002).

[6] Macé, E., et al. Functional ultrasound imaging of the brain: theory and basic principles. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control* 60(3), 492–506 (2013).

[7] Rabut, C., et al. Functional ultrasound imaging of human brain activity through the skull. *Science Translational Medicine* 12(572), eaav4392 (2020).

[8] Rabut, C., Norman, S. L., Griggs, W. S., Russin, J. J., Jann, K., et al. Functional ultrasound imaging of human brain activity through an acoustically transparent cranial window. *Science Translational Medicine* 16, eadj3143 (2024).

[9] Deffieux, T., Demené, C., and Tanter, M. Functional ultrasound imaging: A new imaging modality for neuroscience. *Neuroscience* 474, 110–121 (2021).

[10] Demeulenaere, O., Bertolo, A., Pezet, S., Osmanski, B.-F., Tanter, M., and Deffieux, T. In vivo whole brain microvascular imaging in mice using transcranial 3D ultrasound localization microscopy. *EBioMedicine* 79, 103995 (2022).

[11] Demené, C., et al. Spatiotemporal clutter filtering of ultrafast ultrasound data highly increases Doppler and fUltrasound sensitivity. *IEEE Transactions on Medical Imaging* (2015). DOI: `https://doi.org/10.1109/TMI.2015.2428634`.

[12] Baranger, J., et al. Adaptive spatiotemporal SVD clutter filtering for ultrafast Doppler imaging using similarity of spatial singular vectors. *IEEE Transactions on Medical Imaging* (2018). DOI: `https://doi.org/10.1109/TMI.2018.2789499`.

[13] Yu, A. C. H., and Løvstakken, L. Eigen-based clutter filter design for ultrasound color flow imaging: a review. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control* (2010). DOI: `https://doi.org/10.1109/TUFFC.2010.1521`.

[14] Song, P., et al. Ultrasound small vessel imaging with block-wise adaptive local clutter filtering. *IEEE Transactions on Medical Imaging* (2017). DOI: `https://doi.org/10.1109/TMI.2016.2605819`.

[15] Kasai, C., Namekawa, K., Koyano, A., and Omoto, R. Real-time two-dimensional blood flow imaging using an autocorrelation technique. *IEEE Transactions on Sonics and Ultrasonics* 32(3), 458–464 (1985). DOI: `https://doi.org/10.1109/T-SU.1985.31615`.

[16] Chee, A. J. Y., Yiu, B. Y. S., and Yu, A. C. H. A GPU-parallelized eigen-based clutter filter framework for ultrasound color flow imaging. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control* (2017). DOI: `https://doi.org/10.1109/TUFFC.2016.2606598`.

[17] Lok, U.-W., Tang, J., Kilic, K., Szabo, T. L., and Boas, D. A. Real time SVD-based clutter filtering using randomized SVD and spatial downsampling for microvessel imaging. *Ultrasonics* 107, 106163 (2020). DOI: `https://doi.org/10.1016/j.ultras.2020.106163`.

[18] Demene, C., et al. Improved color Doppler for cerebral blood flow axial velocity estimation (iCD-fUS). *IEEE Transactions on Medical Imaging* 40(5), 1498–1510 (2021).

[19] Tang, J., Kilic, K., Szabo, T. L., and Boas, D. A. Improved color Doppler for cerebral blood flow axial velocity imaging. *IEEE Transactions on Medical Imaging* 40(2), 758–764 (2021).

[20] Demené, C., et al. Functional ultrasound imaging of brain activity in human newborns. *Science Translational Medicine* 9(411), eaah6756 (2017). DOI: https://doi.org/10.1126/scitranslmed.aah6756.

[21] Baranger, J., et al. Bedside functional monitoring of the dynamic brain connectivity in human neonates. *Nature Communications* 12, 1080 (2021). DOI: https://doi.org/10.1038/s41467-021-21387-x.

[22] Soloukey, S., et al. Mobile human brain imaging using functional ultrasound. *Science Advances* 11(25), eadu9133 (2025). DOI: https://doi.org/10.1126/sciadv.adu9133.

[23] Lambert, T., Brunner, C., Urban, A., et al. Functional ultrasound imaging and neuronal activity: How accurate is the spatiotemporal match? *Imaging Neuroscience* 3, IMAG.a.34 (2025). DOI: https://doi.org/10.1162/IMAG.a.34.

[24] Edelman, B. J., et al. The COMBO window: A chronic cranial implant for multiscale circuit interrogation in mice. *PLOS Biology* 22(6), e3002664 (2024). DOI: https://doi.org/10.1371/journal.pbio.3002664.

[25] Chen, Z., et al. Functional ultrasound imaging of auditory responses in comatose patients. *Research* 8, 0709 (2025). DOI: https://doi.org/10.34133/research.0709.

[26] Nunez-Elizalde, A. O., et al. Neural correlates of blood flow measured by ultrasound. *Neuron* 110(10), 1631–1640.e4 (2022). DOI: https://doi.org/10.1016/j.neuron.2022.02.012.

[27] Aaslid, R., Markwalder, T.-M., and Nornes, H. Noninvasive transcranial Doppler ultrasound recording of flow velocity in basal cerebral arteries. *Journal of Neurosurgery* 57(6), 769–774 (1982).

[28] Gao, S., Lam, W. W.-K., Chan, Y.-H., Liu, J., Wong, K.-P., Leung, T. W.-S., et al. Optimal values of flow velocity on transcranial Doppler in grading middle cerebral artery stenosis in comparison with magnetic resonance angiography. *Journal of Neuroimaging* 12(3), 213–218 (2002).

[29] Estrada, H., Gottschalk, S., Reiss, M., Neuschmelting, V., Razansky, D., et al. Observation of guided acoustic waves in a human skull. *Ultrasound in Medicine & Biology* 44(7), 1438–1447 (2018).

[30] Candès, E. J., Li, X., Ma, Y., and Wright, J. Robust principal component analysis? *Journal of the ACM* 58(3), 11 (2011).

[31] Solomon, O., et al. Deep unfolded robust PCA with application to clutter suppression in ultrasound. *IEEE Transactions on Medical Imaging* 39(4), 1051–1063 (2020). DOI: https://doi.org/10.1109/TMI.2019.2941271.

[32] Lin, Z., Chen, M., and Ma, Y. The augmented Lagrange multiplier method for exact recovery of corrupted low-rank matrices. *arXiv preprint* arXiv:1009.5055 (2010).

[33] Halko, N., Martinsson, P. G., and Tropp, J. A. Finding structure with randomness: Probabilistic algorithms for constructing approximate matrix decompositions. *SIAM Review* 53(2), 217–288 (2011).

[34] Soloukey, S., Vincent, A. J. P. E., Smits, M., Slump, C. H., Dirven, C. M. F., et al. Functional ultrasound (fUS) during awake brain surgery: the clinical potential for intra-operative functional and vascular brain mapping. *Frontiers in Neuroscience* 13, 1384 (2019). DOI: https://doi.org/10.3389/fnins.2019.01384.

[35] Shin, Y., et al. Context-Aware Deep Learning Enables High-Efficacy Localization of High Concentration Microbubbles for Super-Resolution Ultrasound Localization Microscopy. Zenodo dataset, DOI 10.5281/zenodo.10711806 (2024).

[36] Shin, Y., et al. Context-Aware Deep Learning Enables High-Efficacy Localization of High Concentration Microbubbles for Super-Resolution Ultrasound Localization Microscopy. *Nature Communications* 15, 3299 (2024). DOI: https://doi.org/10.1038/s41467-024-47154-2.

[37] Chavignon, A., Heiles, B., Hingot, V., Lopez, P., Teston, E., and Couture, O. In vivo rat brain for Ultrasound Localization Microscopy: raw and beamformed data. Zenodo dataset, DOI 10.5281/zenodo.7883227 (2023).

[38] Heiles, B., Chavignon, A., Hingot, V., Lopez, P., Teston, E., and Couture, O. Performance benchmarking of microbubble-localization algorithms for ultrasound localization microscopy. *Nature Biomedical Engineering* (2021). DOI: https://doi.org/10.1038/s41551-021-00824-8.

[39] Errico, C., et al. Ultrafast ultrasound localization microscopy for deep super-resolution vascular imaging. *Nature* 527(7579), 499–502 (2015). DOI: https://doi.org/10.1038/nature16066.

[40] Hingot, V., et al. Subwavelength motion-correction for ultrafast ultrasound localization microscopy. *Ultrasonics* 77, 17–21 (2017). DOI: https://doi.org/10.1016/j.ultras.2017.01.008.

[41] Insana, M. F., Dai, B., Babaei, S., and Abbey, C. K. Combining spatial registration with clutter filtering for power-Doppler imaging in peripheral perfusion applications. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control* 69(12), 3243–3254 (2022). DOI: https://doi.org/10.1109/TUFFC.2022.3211469.

[42] Leonov, D. V., Reshetnikov, R. V., Kulberg, N. S., Nasibullina, A. A., and Gromov, A. I. Doppler Twinkling Artifact Observations: an Open-Access Database of Raw Ultrasonic Signals. *Digital Diagnostics* (2021). DOI: https://doi.org/10.17816/DD76511.

[43] Ozgun, K., and Byram, B. C. Multidimensional clutter filtering of aperture domain data for improved blood flow sensitivity. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control* 68(8), 2645–2656 (2021). DOI: https://doi.org/10.1109/TUFFC.2021.3073292.

[44] Huang, Z., et al. Adaptive higher-order singular value decomposition clutter filter for ultrafast Doppler imaging of coronary flow under non-negligible tissue motion. *Ultrasonics* (2024). DOI: https://doi.org/10.1016/j.ultras.2024.107307.

[45] Garcia, D. SIMUS: an open-source simulator for medical ultrasound imaging. Part I: theory and examples. *Computer Methods and Programs in Biomedicine* 218, 106726 (2022). DOI: https://doi.org/10.1016/j.cmpb.2022.106726.

[46] Cigier, A., Varray, F., and Garcia, D. SIMUS: an open-source simulator for medical ultrasound imaging. Part II: comparison with four simulators. *Computer Methods and Programs in Biomedicine* 220, 106774 (2022). DOI: https://doi.org/10.1016/j.cmpb.2022.106774.

[47] Urban, A., et al. Real-time imaging of brain activity in freely moving rodents using functional ultrasound. *Nature Methods* 12(9), 873–878 (2015). DOI: https://doi.org/10.1038/nmeth.3482.

[48] Demené, C., et al. 4D microvascular imaging based on ultrafast Doppler tomography. *NeuroImage* 127, 472–483 (2016). DOI: https://doi.org/10.1016/j.neuroimage.2015.11.014.

[49] Slepian, D. Prolate spheroidal wave functions, Fourier analysis, and uncertainty—V: The discrete case. *Bell System Technical Journal* 57(5), 1371–1430 (1978). DOI: https://doi.org/10.1002/j.1538-7305.1978.tb02104.x.

[50] Thomson, D. J. Spectrum estimation and harmonic analysis. *Proceedings of the IEEE* 70(9), 1055–1096 (1982). DOI: https://doi.org/10.1109/PROC.1982.12433.

[51] Percival, D. B., and Walden, A. T. *Spectral Analysis for Physical Applications: Multitaper and Conventional Univariate Techniques*. Cambridge University Press (1993).

[52] Guerci, J. R. Artech House monograph (2003).

[53] Wicks, M. C., Rangaswamy, M., Adve, R., and Hale, T. B. *IEEE Signal Processing Magazine* 23(1), 51–65 (2006). DOI: https://doi.org/10.1109/MSP.2006.1593337.

[54] Melvin, W. L., and Guerci, J. R. *IEEE Transactions on Aerospace and Electronic Systems* 42(3), 983–996 (2006). DOI: https://doi.org/10.1109/TAES.2006.248215.

[55] Gurram, P. R., and Goodman, N. A. Spectral-domain covariance estimation with a priori knowledge. *IEEE Transactions on Aerospace and Electronic Systems* 42, 1010–1020 (2006). DOI: https://doi.org/10.1109/TAES.2006.248209.

[56] Van Veen, B. D., and Buckley, K. M. Beamforming: a versatile approach to spatial filtering. *IEEE ASSP Magazine* 5(2), 4–24 (1988). DOI: https://doi.org/10.1109/53.665.

[57] Kuglin, C. D., and Hines, D. C. The phase correlation image alignment method. In *Proceedings of the IEEE International Conference on Cybernetics and Society* (1975).

[58] Stone, H. S., Orchard, M. T., Chang, E.-C., and Martucci, S. A. A fast direct Fourier-based algorithm for subpixel registration of images. *IEEE Transactions on Geoscience and Remote Sensing* 39(10), 2235–2243 (2001). DOI: https://doi.org/10.1109/36.957286.

[59] Foroosh, H., Zerubia, J. B., and Berthod, M. Extension of phase correlation to subpixel registration. *IEEE Transactions on Image Processing* 11(3), 188–200 (2002). DOI: `https://doi.org/10.1109/83.988953`.

[60] Capon, J. High-resolution frequency-wavenumber spectrum analysis. *Proceedings of the IEEE* 57(8), 1408–1418 (1969).

[61] Frost, O. L. An algorithm for linearly constrained adaptive array processing. *Proceedings of the IEEE* 60(8), 926–935 (1972).

[62] Kay, S. M. *Fundamentals of Statistical Signal Processing, Volume II: Detection Theory.* Prentice Hall (1998).

[63] Scharf, L. L., and Friedlander, B. Matched subspace detectors. *IEEE Transactions on Signal Processing* 42(8), 2146–2157 (1994).

[64] Ward, J. MIT Lincoln Laboratory, Tech. Rep. 1015 (1994).